



Converting units of measurement

Teacher resource

Question 1

A rain water tank has a maximum capacity of 8.75 kilolitres.

How many litres could this tank hold when it is full?

875

☐

8 075

☐

8 750

☐

87 500

☐

Answer: C

Skill: Students convert metric units

Background information

The metric system incorporates features of the decimal place value system as well as the consistent use of a set of prefixes. Understanding and working flexibly within the metric system therefore demands an understanding of both decimals and the meanings of the prefixes.

While it is possible to use whole number thinking to make partial sense of the metric system, for example $1\text{kg} = 1000\text{g}$, full flexibility with metric conversions requires decimal understanding. For some students, a lack of understanding of the multiplicative relationships within the decimal system and the associated operational understanding required to meaningfully multiply and divide by multiples of ten, impedes their facility to use the inbuilt features of the metric system.

Additional questions

1. How many grams in 8.75 kilograms?
2. How many kilometres in 8 750 metres?
3. A factory uses 1 000 kilolitres of water. How many litres is this?

Question 2

A cube has edges of 0.5 centimetres.

What is the volume of the cube in cubic millimetres?

- ☐ 5
- ☐ 25
- ☐ 125
- ☐ 500

Answer: C

Skill: Students convert between units of volume

Additional questions

1. A cube has a volume of 0.5 cubic centimetres.
What is this volume in cubic millimetres?
2. A cube has a volume of 0.5 cubic centimetres.
What is this volume in cubic metres

Question 3

Brenda makes raspberry sauce. The mass of 250 millilitres of this sauce is 300 grams.

Brenda makes 3.3 kilograms of sauce. How many litres is this?

_____ litres

Answer: 2.75 litres

Skill: Students use a range of measures in familiar contexts.

Additional questions

1. What is the mass of 4 litres of sauce?
2. Brenda packages her sauce in bottles that each weigh 150g. Each bottle contains 500ml of sauce. What is the mass of one dozen bottles of sauce?

Question 4

Paul buys a 1.25 kilogram tray and a 675 gram tray of mince.

How much mince does Paul buy?

676.25g

☐

800g

☐

1.795kg

☐

1.925kg

☐

Answer: D

Skill: Students calculate and convert familiar metric units

Additional questions

1. Express 4563 grams as kilograms.
2. Express 0.23 kilograms as grams

Student activity: Converting units of measurement

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